AWG19 POLARIZATION DRIVER — CONNECTED SCHEMATIC

NUCLEO-F446RE: see NUCLEO_WIRING.md for cable pin numbers.

U3
LTC1693-5
POL 1
GND 3
GND 4
NC 2
IN
PHASE
GND
NC
VCC 8 +12V
OUT 7 DR4
NC 5
NC 6
R2
22
R5
100k
+12V
Q4
IRF9640
G4
TOP
CLAMP_MID
FAST - TBD2 TVS 43V - TBD
Q4 / Q5 need MCU dead time.
No hardware interlock yet.
Regulated 12 V supply required.
U5
SN74AHCT125PWR
GND 1
POL_3V3 2
GND 4
TG_3V3 5
GND 9
+5V 10
GND 12
+5V 13
10E
1A
2OE
2A
3A
3OE
4A
4OE
1Y 3 POL
2Y 6 TG
3Y 8
4Y 11
VCC 14 +5V
GND 7 GND
J3
NUCLEO-F446RE_CABLE
POL_3V3 1
TG_3V3 2
RX 3
SH 4
DP 5
RESET 6
PA8_/_POL
PA9_/_TG
PB10_/_RX
PB4_/_SH
PB5_/_DP
PA10_/_RESET
PA0_/_I_ADC 7 I_ADC
PC7_/_FF 8 FF
GND 9 GND
3V3_FROM_MCU 10 +3V3
5V_FROM_MCU 11 +5V
GND 12 GND
POL_3V3 R11 100k GND
TG_3V3 R12 100k GND
RX R13 100k GND
SH R14 100k GND
+5V C9 100n / 16V GND
DP R15 100 GND
RESET R16 100 GND
POL R17 10k GND
TG R18 10k GND
J2
EXTERNAL_AWG19
TOP 1
BOTTOM 2
Q5
IRF530
G5
R6
100k
GND
U1
LTC1693-1
POL 1
TG 3
GND 2
GND 4
IN1
IN2
GND1
GND2
VCC1 8 +12V
OUT1 7 DR1
VCC2 6 +12V
OUT2 5 DR5
R1
22
Q1
IRF530
G1
R4
100k
R3
2.2
SENSE_LOW
R9
0.05 / 0.5W
GND
BOTTOM
R7
10k
DAMP
U2A
ADG5412F
BOTTOM 3
PRE 2
S
D
C1
1u
IN
1
RX
U2B
ADG5412F
14
15 GND
IN
16
SH
U2C
ADG5412F
11
10 GND
IN
9
DP
U2D
ADG5412F
6
7 GND
IN
8
RESET
RX_IN
J4
TO_AMPLIFIER
RX_IN 1
GND 2 GND
+3V3
R8
10k
U2E
ADG5412F
+12V 13
−12V 4
VDD
VSS
FF 12
GND 5 GND
FF
U4
INA240A1PW
IN+ 2
IN− 3
+3V3 7
GND 6
REF1
REF2
VS 5 +3V3
OUT 8
GND 4 GND
NC 1
R10
100
CS_OUT
I_ADC
C2
10n
GND
J5
RAW_SHUNT_SENSE
CURRENT_SENSE+ 1 SENSE_LOW
CURRENT_SENSE− 2 GND
J1
12V_INPUT_/_3V3_MONITOR
+12V_REG 1 +12V
−12V 2 −12V
3V3_OUT 3 +3V3
GND 4 GND
U: 3.3V to 5V for gate drivers.
R11–R16: GPIO low when undriven. R17/R18: driver input pulldowns.
Power 3.3V and 5V from Nucleo only. J1-3 is a monitor/output.
No hardware interlock: firmware must prevent POL and TG overlap.
LOCAL BYPASS: place each capacitor at its named IC supply pin
INA240: 1.65 V at zero current: 1 V/A.
J3-7 to PA0 / A0 (single-ended ADC).
J5 raw differential: 50 mV/A.
J3 I_ADC remains amplified.
Polarization current only.
DRAFT — component ratings / footprints still need review.
Current monitor measures polarization, not clamp current.
Kelvin route U4 input pair to R9 pads.
Signal and MCU labels join only the named remote nets.
C3 +12V 1u U1-8 GND
C4 +12V 1u U1-6 GND
C5 +12V 1u U3 GND
C6 +12V 100n U2 GND
C7 −12V 100n U2 GND
C8 +3V3 100n U4 GND

ECE 455 — Jun Yang
Sheet: /
File: Polarization_Nucleo_F446RE_RevE.kicad_sch
Title: AWG19 Polarization — Connected Circuit
Size: A3 | Date: | Rev: DRAFT E
KiCad E.D.A. 10.0.6 | Id: 1/1